

A Novel Fault Detection and Exclusion Method for Applying Low-Cost INS/GNSS Integrated Navigation System in Urban Environments

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Abstract—Achieving higher accuracy in positioning at lower cost in urban environments is a fundamental requirement in modern intelligent transportation, autonomous driving, etc. The integration of low-cost inertial navigation system (INS) and global navigation satellite system (GNSS) is a promising choice to meet this objective, however, it still requires a suitable fault detection and correction method. Inspired by the structure of multi-channel parallelism in vector tracking, this paper introduces a multi-channel parallel processing (MCP) method for identifying and eliminating GNSS faulty signals in the measurement domain in tightly integrated structure, aided by the principle of multiple model interaction. At each positioning epoch in each channel, the MCP method individually compares the difference between each satellite's measurement and its INS estimation version. These differences are normalized by a multiple model interaction process to derive the probability that each satellite is error free. These probabilities, linked to each satellite, can be used as weights for the optimal output or as indicators to identify false signals. The road test results in the Zhongguancun area of Beijing demonstrate that the method's performance is comparable to that of state-of-the-art methods and avoids the unwanted outcome of selecting a fixed error elimination threshold that leads the system to diverge in a distinct environment or positioning mode. The results were valid in both tightly coupled (TC) and weighted least square (WLS) navigation modes, suggesting that the method can be utilized for single vehicle as well as vehicle-to-vehicle or vehicle-to-road cooperative positioning.

Index Terms—Urban environments, fault detection and exclusion, INS/GNSS integration.

I. INTRODUCTION

ACCURATE and reliable positioning results are an essential requirement in many modern urban applications such as autonomous driving, intelligent transport, etc. [1] and [2]. Like any technology, a navigation system can occasionally produce output errors that are much larger than the specified uncertainty limits due to hardware or software failures. To get the best performance out of a navigation system, fault detection and exclusion (FDE) techniques should be implemented. GNSS failure modes may be divided into four categories: satellite faults; unusual atmospheric propagation; local channel

failures, and general user equipment faults. In open environments, the use of ground-based signal monitoring networks and redundant hardware can handle most failures. However, urban environments are high rejection scenarios for GNSS where facilities such as tall buildings, trees, bridges, etc., can be found everywhere. They block and reflect GNSS signals, causing non-line-of-sight (NLOS) reception, multipath effects, signal outliers and signal outages that are closely related to the user's location, resulting in local channel failures [3], [4], [5]. Although the inclusion of other sensors (usually with INS) allows the system to overcome the problem of poor satellite configuration in urban environments, NLOS reception and multipath effects can still significantly degrade the performance of the system due to the dominance of GNSS in the integration [6]. Therefore, if faulty GNSS signals caused by NLOS or multipath can be eliminated, the positioning performance of the system will be greatly improved. It is also a prerequisite for providing a real-time measure of the level of trust to be placed in the localization estimates as vehicles operate [7].

In order to fix the performance of GNSS in degraded environments, a lot of research has been carried out on FDE methods [8]. The original FDE algorithms were developed for aviation, hence many of the FDE algorithms cannot be directly applied to dynamic applications such as transportation in urban areas. Additionally, the urban environment is flooded with reflected signals that can cause large and rapidly changing errors compared to the aviation environment, reducing measurement redundancy, and increasing the possibility of multiple faults occurring at the same time [9]. Among the FDE algorithms applied in urban environments, the method of weighting measurements is the most straightforward and widely used class. This class of methods weights the measurements based on information such as the elevation angle of the satellite, the signal-to-noise ratio of the signal, etc., providing the final positioning result using the least squares method [10]. However, this kind of information cannot be used as a basis for judging whether the signal is good or bad, since the signal-to-noise ratio of the reflected signal is sometimes even higher than that of the direct signal [11]. Receiver autonomous integrity monitoring (RAIM) is another widely used FDE method. RAIM-like methods use least squares residuals to perform consistency checks by traversing the positioning results of a subset of the satellites available at the current moment, enabling the elimination of errors [12], [13]. The method

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requires that the number of visible satellites must exceed six and fails when multiple contaminated measurements are encountered at the same time (which can be frequent in urban environments).

In recent years, with the development of computer technology and communication technology, people introduce artificial intelligence, 3-D map assistance, and multi-vehicle cooperative positioning into the cutting-edge FDE methods. The AI-based approaches classify GNSS signals using features extracted from GNSS measurements to distinguish between direct, multipath and non-line-of-sight signals [14], [15]. AI-type methods can achieve good results when sufficient data and training time are available, but AI-type methods are environmentally sensitive and work under the assumption that the noise characteristics are consistent. Their relevance can only be satisfied when the external conditions are constant [16]. The 3-D map-assisted technique is based on the fact that the reflection of GNSS signals is mainly generated by building surfaces, and the ray tracing method is used to simulate the signal reflections from all building surfaces after 3-D modelling of the buildings in the city to find out the reflected signals [17], [18]. Though methods have begun to be explored to lighten the computational load of 3-D map-assisted technology, it is still computationally intensive and as environmentally determined as AI-type approaches [17], [18]. Multi-vehicle cooperative technology assumes that GNSS measurements and in-car ranging values can be shared between multiple vehicles [20]. It can actually be seen as a mobile RTK network. When applying FDE using multi-vehicle cooperation techniques, the principle is the same as the classical RAIM method, except that the performance is increased by the more information redundancy due to multiple vehicles [21], [22]. Real-time synchronization of ranging information between multiple vehicles is an unsolvable problem for the time being.

Since more information redundancy would be more helpful for FDE implementation, it can also be supplemented by adding historical information or additional sensors [23], [24]. The historical information from a single system is not so valuable compared to the reliable additional information from additional sensors. At the same time, the method of enhancing the information redundancy through additional sensors is also more economical and practical than the state-of-the-art solutions mentioned earlier. Especially, the INS/GNSS integrated navigation system has been a hotspot for research and application for decades by the advantage of the complementary error characteristics of INS and GNSS and with the gradual maturity of low-cost inertial measurement units (IMUs), the popularity of INS/GNSS integrated navigation systems is increasing year by year. To this end, FDE approaches using multiple MEMS-IMUs have been investigated [25], and the FDE algorithms performance differences between loosely and tightly INS/GNSS integrated mode have been analyzed respectively [26] and [27]. A multi-failure FDE algorithm based on pseudorange comparison is used in on-board INS/GNSS integrated navigation [9], but it needs initial database generation. An urban INS/GNSS integrated navigation quality control algorithm based on the dual w-test is proposed, which requires

sufficient visible satellites, otherwise the double-w test cannot be performed [28]. In the INS/GNSS tight integration, the innovation filtering method normalizes the measurement innovation (pseudorange and pseudorange rate) of all satellites to exclude quantities exceeding a threshold, also achieves good results [29]. These FDEs mentioned above do not take into account the extent of the error when they are executed [30]. They eliminate the corresponding signals once the error exceeds a certain threshold, while the signals below the threshold are considered to be of equal importance. Even if a signal containing a glitch reduces positioning accuracy, it may still contain useful information [31], and different signals should have different effects on positioning accuracy even if thresholds are not exceeded. Hence, the conventional FDE algorithm may cause the system to lock on to the wrong value and eventually lead to filter divergence when multiple errors are encountered at the same time. An adaptive fault-tolerant algorithm based on the innovation sequence is proposed to improve relative navigation accuracy by isolating the covariance estimation and adaptively adjusting the corresponding measurement noise covariance matrix of minor faults signals [31]. Its performance is very dependent on the accuracy of the priori stochastic information. By adjusting the corresponding R-matrix, the method is limited in its adjustment effort, measurements with large errors can only be excluded, otherwise they may still lead to filter dispersion, and it still performs the error detection part based on the threshold. Therefore, it is necessary to develop an FDE algorithm that is tolerant to errors, can be used when the number of satellites is small, and does not depend on a priori information.

Based on the above analysis, it can be found that enhancing the FDE algorithm by improving the redundant information utilization can avoid the complexity and time synchronization problems brought by multi-sensors or multi-vehicles while obtaining higher performance.

To this end, we have made the following contributions:

- (a) An improved parallel FDE method based on tightly coupled structures is proposed (i.e. the MCPP method). The method borrows the idea of multi-channel parallel operation in vector tracking, exploits the single-channel positionable property of tightly-coupled structure, and combines the interacting multiple model (IMM) algorithm. The method can distinguish the contribution degree of each satellite, which further improves the information utilization of the system.
- (b) The advantages and limitations of the proposed fault-tolerant method are analyzed through experiments in Zhongguancun under the TC mode, and the feasibility of trading accuracy for performance is explored.
- (c) The performance of multiple methods was compared in WLS positioning mode to further validate the performance of the MCPP method being used for lead vehicles or road facilities in cooperative positioning.

To be specific, the proposed MCPP FDE method inputs measurements (pseudorange and pseudorange rate) from each satellite into separate channels (sub-filters) for calculation. During the calculation, in each sub-model, the difference

between the measurement of a satellite and its INS estimate version is compared. These differences are normalized by a multiple model interaction process to derive the probability that each satellite is error free. These probabilities corresponding to each satellite can be used as weights for the optimal output or as measures for identifying false signals. So, the method can switch between two FDE modes, fault-tolerant and fault-exclusion, which are corresponding to MCPP-weighted mode and MCPP-exclusion mode respectively in this paper. In which the MCPP-exclusion mode is similar to the existing threshold-based FDE methods, and its performance mainly depends on how accurately the system identifies the error at the current epoch, and the best performance can be obtained if the identification is accurate. In the MCPP weighted mode, the degree of error in the signal is taken into account, preserving a portion of the useful information while reducing the impact of the error. The correct measurements can also be given different weights depending on how much they contribute to the positioning results, allowing for better results. As this mode still retains some of the fault information causes the method to be inferior to threshold-based methods in the extreme conditions, but it also makes the method more tolerant to errors. In contrast to the threshold-based FDE method, it does not require an artificially configured threshold, which avoids the situation where the system sometimes locks onto the wrong measurements and eventually diverges due to the accumulation of errors.

The IMM algorithm used in this paper originated as a modular target tracking method that was widely used in the 1990s for aircraft sensor fault detection and estimation [32]. In the field of navigation, the IMM algorithm is often used in various strongly nonlinear scenarios to achieve a better robustness to highly dynamic or model mismatching. In existing work, IMM algorithms are mostly used to decompose the complex behavior of the target into a superposition of multiple simple behaviors (sub-models) and compute their respective weights at each positioning epoch based on the output of each sub-model [33]. For example, multiple noise models are used to solve the uncertainty of navigation filter parameters [34], [35], and multiple motion models such as constant velocity, constant steering, and constant acceleration are used to cover the motion states [36], [37].

An IMM-EKF method is proposed for the fusion filtering of inertial, odometry, and satellite data [38]. The implementation of IMM-based methods allows the possibility of using highly dynamic models just when required, diminishing unrealistic noise considerations in non-maneuvering situations, and the computational charge of the system. Based on similar considerations, [39] proposes an IMM-based method for predicting lane changes in highways, and studied different model sets to meet the needs of the IMM-based algorithm. An adaptive IMM algorithm is achieved by adjusting the probability transfer matrix in real time [40]. It is used to associate the Huber's M-estimation based robust filter with the RAIM filter, to take advantage of both filters in a complementary sense. Although IMM methods have been widely used in navigation, they can be further combined with other algorithms to release greater potential.

With the establishment of low earth orbit satellite constellations and the explosion in the number of available satellites in future, the threshold-based FDE method will be more likely to lock on to the wrong signals, leading to signal dispersion, which is not a problem with the MCPP-weighted mode. Moreover, the results obtained by MCPP method can be directly used for WLS positioning without extra costs. In future intelligent transport scenarios such as vehicle-road and vehicle-vehicle cooperation, it can be used as an augmented message broadcasted to the surroundings by the head vehicle or road facility, thus improving the performance of the intelligent transport system.

The rest is organized as follows. The overview of the proposed method will be introduced in Section II. In Section III, the construction of the system model, the state selection, the multiple model (channel) interaction and the system workflow will be shown, respectively. In Section IV, the experimental scenarios and results will be introduced, and the analysis will be discussed. Conclusions and future work can be found in Section V.

II. SYSTEM OVERVIEW

The overall structure of our system is shown in Fig. 1. This newly FDE method takes advantages of the IMM algorithm, which allows multiple models (channels) to work in parallel and output optimal signal integrity assessment results. In this method, the measurements corresponding to each satellite are fed separately into individual parallel tightly integrated navigation models, which are then fused using the principle of multiple model interaction. The measurement values of GNSS system, the estimated values derived from the output of INS system, and the input interaction among parallel tightly coupled filtering results corresponding to each satellite are jointly used as the outputs of the system propagation phase of the novel FDE method. The results of each tightly coupled model, the probabilistic update results and the navigation processor results are used as the outputs of the measurement update phase. At each positioning epoch, the system evaluates the weights of each tightly coupled model based on their innovation and innovation covariance matrix to obtain the optimal signal integrity assessment results and then outputs the positioning results for different positioning modes as required. As shown in Fig. 1, the method consists of the following steps:

1. Calculate the estimated pseudorange and pseudorange rate $\hat{\rho}^{(n)}, \hat{\dot{\rho}}^{(n)}$ at the current epoch based on the ephemeris data and the state value output from the INS system.
2. Calculate the measurement innovation of the system $\delta\hat{\rho}^{(n)}, \delta\hat{\dot{\rho}}^{(n)}$ at the current epoch based on the estimated value in step 1 and the measurement value of the GNSS system.
3. Calculate the mixture states estimate $\hat{x}_{oj,k-1}$ and their error covariance matrix $P_{oj,k-1}$ for the current epoch based on the output values of each model, the probabilities of each model and the positioning results of system at the previous epoch.
4. Calculate the states estimate $\hat{x}_{i,k}$ and their error covariance matrix $P_{i,k}$ corresponding to each satellite based on the $\delta\hat{\rho}^{(n)}, \delta\hat{\dot{\rho}}^{(n)}$ from step 2, the $\hat{x}_{oj,k-1}$ and $P_{oj,k-1}$ from step 3, and the positioning results of system at the previous epoch.

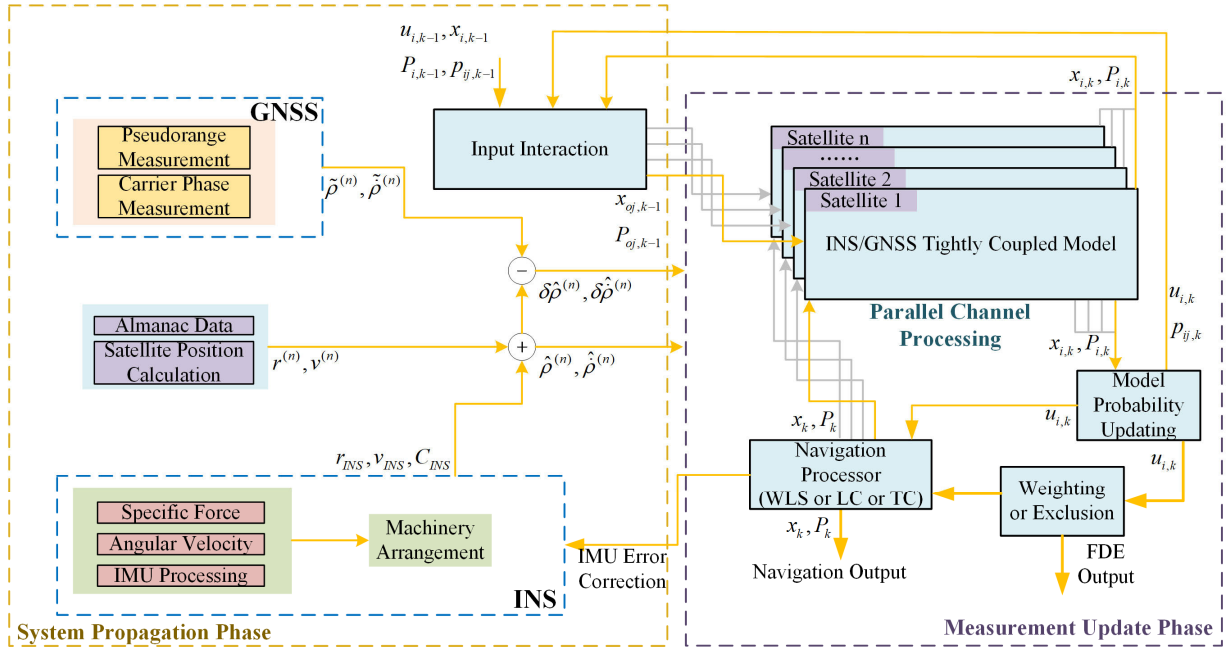


Fig. 1. General structure of the system.

5. Calculate the probability of each model based on the $\delta\hat{\rho}^{(n)}$, $\delta\hat{\rho}^{(n)}$ from step 2, and the $\hat{x}_{i,k}$ and $P_{i,k}$ from step 4 to output the FDE results.

6. Calculate the navigation result x_k based on the GNSS measurement values, INS measurement values and the FDE results.

The principles underlying the method are presented in the next section.

III. MULTI-CHANNEL PARALLEL PROCESSING

A. System Model and State Selection

To make the method capable of processing satellite signals in parallel, it needs to rely on a tightly coupled INS/GNSS structure. Since the tight structure works by using the output of the GNSS to correct the error of the INS, the state equation and state transition equation of the integrated system are derived from the error propagation equation of INS [29]. For tightly integrated mode, in the ECEF coordinate system, its state equation can be expressed as:

$$\dot{x} = Fx + \omega. \quad (1)$$

where, x represents the state quantity, F represents the system matrix, and ω represents the system noise. The state variables of a tightly integrated navigation system x generally have 17 dimensions, including five 3-dimensional state variables and two 1-dimensional local clock error parameters. The five 3-dimensional state variables include: position error δr , velocity error δv , attitude error $\delta\psi$, accelerometer bias error b_a , and gyroscope bias error b_g . The local clock error state parameter is $\delta\rho_c$ and $\delta\dot{\rho}_c$. It can be expressed as:

$$x = [\delta r \quad \delta v \quad \delta\psi \quad b_a \quad b_g \quad \delta\rho_c \quad \delta\dot{\rho}_c]^T. \quad (2)$$

Based on the INS error propagation equation and the state quantities, the system matrix can be expressed as (3), shown at

the bottom of the next page, where γ_{ib}^e is the gravitational term proportional to the local altitude, r_{eb}^e is the local position, r_{es}^e is the local geocentric radius, ω_e^e is the angular rate of earth rotation, $[\omega_e^e \wedge]$ is the skew symmetric matrix of earth rate, C_b^e is the coordinate transformation matrix from the native to the ECEF coordinate system, and f_{ib}^b is the specific force measured by INS. When the system state propagation interval is τ_s , the state transfer matrix of the system Φ can be expressed as:

$$\Phi = \sum_{n=0}^{\infty} \left(\frac{F^n \tau_s^n}{n!} \right). \quad (4)$$

B. Multiple Channel Interaction Algorithm

The method in this paper takes advantages of the IMM algorithm and the tightly coupled structure. The IMM algorithm is used to process multiple sub-KFs in parallel. In the original IMM algorithm, each sub-filter corresponds to a different state model, and different state models describe different motions, so each sub-filter has a different estimate of the target state. In each sub-filter of each filtering epoch, the initial conditions corresponding to the specific model are computed by mixing the state estimates of all sub-filters of the previous epoch, and then the regular filtering step is performed. Finally, the model probabilities are updated based on the model matching likelihood function, thus weighting the output of the state estimates of all sub-filters.

The multi-model interaction algorithm in this paper is based on the IMM algorithm and consists of four steps: input interaction, model parallel filtering, model probability update, and navigation processing. Slight improvements have been made to the algorithm to enable its use in GNSS FDE. In this paper, the model update probability here can be used as the

FDE result and the optimal navigation estimate at the current moment is calculated based on the FDE result.

1) *Input Interaction*: Compute the mixture state estimate $\hat{x}_{oj,k-1}$ and its covariance matrix $P_{oj,k-1}$ for observation model j at moment $k-1$.

$$\begin{aligned}\hat{x}_{oj,k-1} &= \sum_{i=1}^n \hat{x}_{i,k-1} \cdot u_{ij,k-1} \\ P_{oj,k-1} &= \sum_{i=1}^n u_{ij,k-1} \\ &\quad \times \{P_{i,k-1} + [x_{i,k-1} - \hat{x}_{oj,k-1}][x_{i,k-1} - \hat{x}_{oj,k-1}]^T\},\end{aligned}\quad (5)$$

where n denotes the number of models corresponding to the number of satellite, $\hat{x}_{i,k-1}$ is the state vector of the i -th model at moment $k-1$, $P_{i,k-1}$ is the error covariance matrix of the i -th model at moment $k-1$, and $u_{ij,k-1}$ is the mixture model probability from model i to model j at moment $k-1$, which can be expressed as:

$$u_{ij,k-1} = \frac{p_{ij,k-1} u_{i,k-1}}{\sum_{i=1}^n p_{ij,k-1} u_{i,k-1}}, \quad (7)$$

where $p_{ij,k-1}$ is an element of the probability transfer matrix Π_{k-1} . In traditional IMM algorithms, a fixed Markov probability transfer matrix is used to determine the transition relationship between models, where the element p_{ij} represents the probability of transferring from the i -th model to the j -th model. Since the probability of an error occurring in satellite is very low, and with the help of tightly integrated structures, GNSS only provides pseudo-range measurements and does not participate in the positioning. Thus, the system is free from the constraints of factors such as satellite elevation angle. Therefore, all the satellites should have the same weight with respect to each other and for each moment there is:

$$p_{ij,k} = \frac{1}{n}. \quad (8)$$

The probability transfer matrix Π_k is:

$$\Pi_k = \begin{pmatrix} p_{11} & \cdots & p_{1n} \\ \vdots & \ddots & \vdots \\ p_{n1} & \cdots & p_{nn} \end{pmatrix}_k = \begin{pmatrix} \frac{1}{n} & \cdots & \frac{1}{n} \\ \vdots & \ddots & \vdots \\ \frac{1}{n} & \cdots & \frac{1}{n} \end{pmatrix}_k. \quad (9)$$

2) *Model Parallel Filtering*: In the method, for every single model, the observations used are the pseudorange $\tilde{\rho}^{(n)}$ and pseudorange rate $\tilde{\dot{\rho}}^{(n)}$ measurements corresponding to each

satellite output by the GNSS ranging processor. At moment k , for the s -th satellite there is:

$$\tilde{z}_k^s = [\tilde{\rho}_k^s, \tilde{\dot{\rho}}_k^s]^T. \quad (10)$$

The state estimate of each filter, $\hat{x}_{i,k/k-1}$, is equal to the output of INS, while the i -th error covariance matrix, $P_{i,k/k-1}$, can be expressed as:

$$P_{i,k/k-1} = \Phi_i P_{oj,k-1} \Phi_i^T + Q_i. \quad (11)$$

The system's predicted pseudorange $\hat{\rho}^{(n)}$, pseudorange rate $\hat{\dot{\rho}}^{(n)}$ and line-of-sight vector $u^{(n)}$ for each satellite can be obtained based on the corrected inertial navigation output, the estimated local clock offset and clock drift, and the satellite positions and velocities calculated from the navigation messages. For the s -th satellite, the observation equation and observation matrix of the system at time k , H_k^s , can be expressed as:

$$\begin{aligned}z_k^s &= H_k^s \hat{x}_{i,k/k-1} + v = [\hat{\rho}_k^s, \hat{\dot{\rho}}_k^s]^T + v \\ H_k^s &= \begin{pmatrix} u_k^{sT} & 0_{1 \times 3} & 0_{1 \times 3} & 0_{1 \times 3} & 0_{1 \times 3} & 1 & 0 \\ 0_{1 \times 3} & u_k^{sT} & 0_{1 \times 3} & 0_{1 \times 3} & 0_{1 \times 3} & 0 & 1 \end{pmatrix},\end{aligned}\quad (12)$$

where v represents measurement noise. At this point, the system innovation δz_k^s is the difference between the observed and predicted values of pseudorange and pseudorange rate:

$$\delta z_k^s = [\delta \hat{\rho}_k^s, \delta \hat{\dot{\rho}}_k^s]^T. \quad (14)$$

At time k , the measurement innovation of the s -th satellite can be represented as:

$$\begin{aligned}\delta \hat{\rho}_k^s &= \tilde{\rho}_k^s - \hat{\rho}_k^s \\ \delta \hat{\dot{\rho}}_k^s &= \tilde{\dot{\rho}}_k^s - \hat{\dot{\rho}}_k^s.\end{aligned}\quad (15)$$

At time k , the state vector update equation of the s -th satellite can be expressed as:

$$\begin{aligned}\hat{x}_{i,k} &= \hat{x}_{i,k/k-1} + K_{i,k}(\tilde{z}_k^s - H_k^s \hat{x}_{i,k/k-1}) \\ &= \hat{x}_{i,k/k-1} + K_{i,k} \begin{pmatrix} \delta \hat{\rho}_k^s \\ \delta \hat{\dot{\rho}}_k^s \end{pmatrix}\end{aligned}\quad (16)$$

$$P_{i,k} = [I - K_{i,k} H_k^s] P_{i,k/k-1}. \quad (17)$$

3) *Model Probability Update*: The maximum likelihood function of each model at moment k is used as the basis for judging the degree of similarity between the corresponding pseudorange estimates and pseudorange measurements for each satellite, and the normalized result is used as the weight

$$F = \begin{pmatrix} 0_{3 \times 3} & I_{3 \times 3} & 0_{3 \times 3} & 0_{3 \times 3} & 0_{3 \times 3} & 0_{3 \times 1} & 0_{3 \times 1} \\ -\frac{2\gamma_{ib}^e}{r_{es}^e} r_{eb}^{eT} & -2[\omega_{ie}^e \wedge] & -(C_b^e J_{ib}^b) \wedge & C_b^e & 0_{3 \times 3} & 0_{3 \times 1} & 0_{3 \times 1} \\ 0_{3 \times 3} & 0_{3 \times 3} & -[\omega_{ie}^e \wedge] & 0_{3 \times 3} & C_b^e & 0_{3 \times 1} & 0_{3 \times 1} \\ 0_{3 \times 3} & 0_{3 \times 3} & 0_{3 \times 3} & 0_{3 \times 3} & 0_{3 \times 3} & 0_{3 \times 1} & 0_{3 \times 1} \\ 0_{3 \times 3} & 0_{3 \times 3} & 0_{3 \times 3} & 0_{3 \times 3} & 0_{3 \times 3} & 0_{3 \times 1} & 0_{3 \times 1} \\ 0_{1 \times 3} & 0_{1 \times 3} & 0_{1 \times 3} & 0_{1 \times 3} & 0_{1 \times 3} & 0 & 0 \\ 0_{1 \times 3} & 0_{1 \times 3} & 0_{1 \times 3} & 0_{1 \times 3} & 0_{1 \times 3} & 0 & 0 \end{pmatrix}. \quad (3)$$

of each model at this moment. The maximum likelihood function that best matches model i at moment k is:

$$\Lambda_{i,k} = \frac{1}{(2\pi)^{\frac{1}{2}} |S_{i,k}|^{\frac{1}{2}}} \exp \left\{ -\frac{1}{2} d_{i,k}^T S_{i,k}^{-1} d_{i,k} \right\}, \quad (18)$$

where $d_{i,k}$ is the innovation vector and $S_{i,k}$ is the innovation covariance matrix. They can be expressed as:

$$d_{i,k} = \tilde{z}_k^s - H_k^s \hat{x}_{i,k/k-1} \quad (19)$$

$$S_{i,k} = H_k^s P_{i,k} H_k^{sT} + R_{i,k}, \quad (20)$$

where R_i is the covariance matrix of the observed noise v_j . Its pseudorange and pseudorange rate noise term are $100 \text{ (m}^2\text{)}$ and $1 \text{ (m}^2\text{/s}^2\text{)}$ respectively. The update probability of model i is:

$$u_{i,k} = \frac{\Lambda_{i,k} \sum_{j=1}^n p_{ij} u_{j,k-1}}{\sum_{i=1}^n \sum_{j=1}^n \Lambda_{i,k} p_{ij} u_{j,k-1}}. \quad (21)$$

4) *Navigation Processing*: The final output at moment k is obtained from the filtering results and its model probabilistic. For tightly coupled navigation:

$$\hat{x}_k = \sum_{i=1}^n \hat{x}_{i,k} u_{i,k} \quad (22)$$

$$P_k = \sum_{j=1}^n u_{j,k} \{ P_{j,k} + [\hat{x}_{j,k} - \hat{x}_k][\hat{x}_{j,k} - \hat{x}_k]^T \}. \quad (23)$$

For loosely coupled navigation and weighted least square navigation, the model probabilistic can be used as the weight in the navigation solving equation:

$$\begin{aligned} \hat{x}_{k,GNSS} &= \hat{x}_{k-1,GNSS} \\ &+ \left(H_{k,GNSS}^T C^{-1} H_{k,GNSS} \right)^{-1} H_{k,GNSS}^T C^{-1} \delta z_k \end{aligned} \quad (24)$$

$$C = u_k^{(n)}. \quad (25)$$

C. System Workflow

Combining all of the above, the main algorithmic workflow of the system is shown in the table below:

IV. EXPERIMENTS AND DISCUSSIONS

A. Experimental Scenario and Setup

We conducted field tests in the Zhongguancun area of Beijing. Our driving trajectory and the failure situation of the GNSS signal along the trajectory are shown in Fig. 2. The fault situation of the GNSS signal in the figure is obtained by normalizing the positioning error results of the pure GNSS system. The normalization of the positioning error in the test was carried out by taking the minimum value of the positioning error in the final experimental results as 1 and the maximum value of the positioning error as 10.

Since the systems operating in the real world contain many unobservable errors, and there is no way to determine the true

Algorithm 1 MCPP

Input: $x_0, P_0, p_{ij}, u_0, R_0, Q_0, \Phi$

(1) System propagation phase:

for $i = 1: n$

$$u_{ij,k-1} = \frac{p_{ij,k-1} u_{i,k-1}}{\sum_{i=1}^n p_{ij,k-1} u_{i,k-1}}$$

$$\hat{x}_{oj,k-1} = \sum_{i=1}^n \hat{x}_{i,k-1} \cdot u_{ij,k-1}$$

$$P_{oj,k-1} = \sum_{i=1}^n u_{ij,k-1} \{ P_{i,k-1} + [\hat{x}_{i,k-1} - \hat{x}_{oj,k-1}][\hat{x}_{i,k-1} - \hat{x}_{oj,k-1}]^T \}$$

$$\hat{x}_{i,k/k-1} = x_{k,INS}$$

$$P_{i,k/k-1} = \Phi_i P_{oj,k-1} \Phi_i^T + Q_i$$

end

(2) Measurement update phase:

for $i = 1: n$

$$K_{i,k} = P_{oj,k-1} H_k^{sT} [H_k^s P_{oj,k-1} H_k^{sT} + R_{i,k}]^{-1}$$

$$\hat{x}_{i,k} = \hat{x}_{i,k/k-1} + K_{i,k} (\tilde{z}_k^s - H_k^s \hat{x}_{i,k/k-1})$$

$$P_{i,k} = [I - K_{i,k} H_k^s] P_{i,k/k-1}$$

$$S_{i,k} = H_k^s P_{i,k} H_k^{sT} + R_{i,k}$$

$$d_{i,k} = \tilde{z}_k^s - H_k^s \hat{x}_{i,k/k-1}$$

$$\Lambda_{i,k} = \frac{1}{(2\pi)^{\frac{1}{2}} |S_{i,k}|^{\frac{1}{2}}} \exp \left\{ -\frac{1}{2} d_{i,k}^T S_{i,k}^{-1} d_{i,k} \right\}$$

$$u_{i,k} = \frac{\Lambda_{i,k} \sum_{j=1}^n p_{ij} u_{j,k-1}}{\sum_{i=1}^n \sum_{j=1}^n \Lambda_{i,k} p_{ij} u_{j,k-1}}$$

end

As for MCPP-weighted:

$$\hat{x}_k = \sum_{i=1}^n \hat{x}_{i,k} u_{i,k}$$

As for MCPP-exclusion:

Exclude $\hat{x}_{i,k}$ when $u_{i,k} < \text{threshold}$

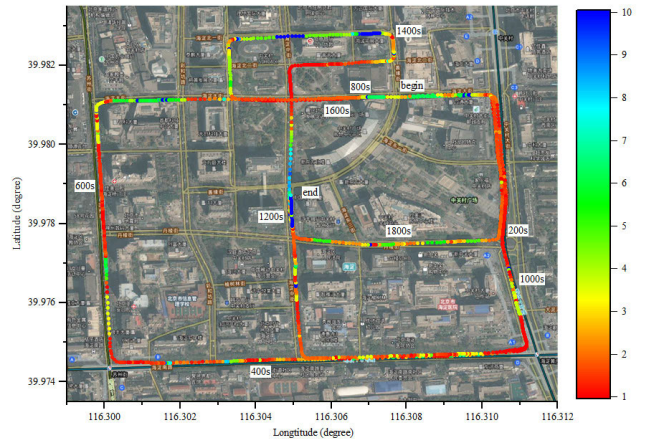


Fig. 2. Trajectories and distribution of contamination of GNSS signals, where the colors represent the normalized results of the accuracy of GNSS positioning results here, with the least contamination in red. The fault situation of the signal in the figure includes both the outlier and reflection effects (multipath, non-line of sight) of GNSS system in urban environment.

value. The localization result of the system is just an optimal estimate of the solution based on these error-containing signals by some algorithm (e.g., iterative least squares). The errors caused by multipath effects and NLOS reception are strongly correlated with the location and poorly observable, they can

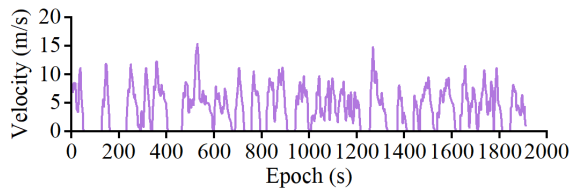


Fig. 3. The distribution of vehicle speed along time, which reflects the maneuvering of the vehicle.

only be identified through consistency checks in terms of their contribution to the equivalent distance error. So, the evaluation of whether a signal is faulty (or anomalous) therefore depends on how much it contributes to the positioning error, and is ultimately reflected in the discrepancy between the positioning result based on the signal and the consistency. Signals that significantly increase the error in the localization results are considered to be faulty and the signal failure situation of GNSS in Fig. 2 includes both the outlier and reflection signals that are prevalent in the urban environment.

The time tag in the figure represents the time elapsed from the departure time when travelling to this point. Fig. 3 shows the vehicle speed along time, which can reflect the maneuvering of the vehicle. It can be seen from the graph that we experienced frequent maneuvers during our drive. Combining the impact on the GNSS system reflected in Fig. 2, we can see that we have chosen a typical driving scenario in an urban environment.

Two integrated navigation receivers are installed in the same car, with reference trajectories provided by the SPAN-ISA-100C, which integrates a GNSS receiver PwrPak7 from NovAtel and an inertial measurement unit IMU-ISA-100C from Norhrop-Grumman Litef, equipped with a fiber optic gyroscope. With post-processing mode, its nominal position accuracy (RMS) in the horizontal direction can reach 0.010 m. The second device is the low-cost Npos 220, which is equipped with a MEMS-IMU whose bias stability and random walk of the accelerometer at the level of 10 mg and $100 \mu g/\sqrt{Hz}$, and bias stability and random walk of the gyroscope at the level of $10^\circ/h$ and $1^\circ/\sqrt{h}$ respectively.

B. Performance Analysis of the Multi-Channel Parallel Processing Method

A total of 6 FDE methods were selected for comparison and validation in the above scenarios:

- INS/GNSS tightly coupled mode without FDE method.
- MCPP-weighted method. The method uses the weights of the satellite measurements output by the MCPP filter to get an optimal FDE result.
- MCPP-exclusion method. The method detects and excludes faults below a certain weighting threshold.
- Cooperative reliability message (CRM) based method. A state of art FDE method that uses the quality of GNSS measurements offered by nearby cooperators as the prior knowledge together with the consistency check on raw GNSS measurements to get an improved FDE result [22].

- Innovation based method. A powerful classical FDE method that normalizes the innovation of positioning filter and excludes faults below a certain threshold [29].
- Classical RAIM method.

Among them, the CRM and RAIM purely use GNSS information, in contrast to MCPP-weighted, MCPP-exclusion and innovation-based methods that also use INS measurements. The CRM, 2 kinds of MCPP and innovation-based methods enhance the redundancy of system information by different means, while RAIM uses only existing redundant information. All thresholds (if there is) above were obtained by exhaustive examination method based on the mean positioning error. The results of their identification of false signals are posted in the appendix.

Fig. 4 takes the signals of satellite b6 and satellite b21 as examples to show the fault signal recognition results of several methods at each epoch. The positioning error statistics for these methods in TC positioning mode are shown in Table I. Their distributed positioning performance along trajectory and time in TC positioning mode are shown in Fig. 5 and Fig. 6 respectively. Their positioning error cumulative distribution function (CDF) are shown in Fig. 7.

In Fig. 4, the MCPP-weighted mode gives the normalized probability that the method considers the satellite signal to be error-free in each epoch. The normalization process and indicators are shown in section III, part B, 3). The sum of the error-free probabilities of all satellites in each epoch is 1. The rest are all threshold-based exclusion methods. Each red square in each figure indicates that the signal is considered to be faulty by the method at its corresponding epoch.

On the premise of ensuring that the number of available satellites meets the positioning requirements, combined with the results in Fig. 5, Fig. 6 and Table I, it can be roughly believed that when the method based on the consistency check principle is working normally, the stricter the fault elimination is, the smaller the average positioning error of the system. However, subsequent experiments have also demonstrated that stricter fault elimination thresholds lead to an increased probability of divergent results.

More fault identification results are shown in the appendix. The attached Fig. 15 shows the original satellite signal visibility, where the symbol 1 indicates that the corresponding satellite signal is received at the epoch, and the symbol 0 indicates that the corresponding satellite signal is not received at the epoch. The attached Fig. 16 ~ Fig. 20 show the fault identification of each satellite signal by several methods at each epoch.

As can be seen from the statistics of the average and the standard deviation of the positioning error in Table I, compared to the localization performance when no FDE is used, each FDE method is able to improve the localization performance of the system, indicating that they all play their respective effects. The CRM, 2 kinds of MCPP and innovation-based methods which enhance the information redundancy of the system have better results. In contrast, RAIM methods have relatively poor results. The maximum error of the RAIM method is even higher than that of the normal TC results without FDE. The insufficient performance of the RAIM method compared to the

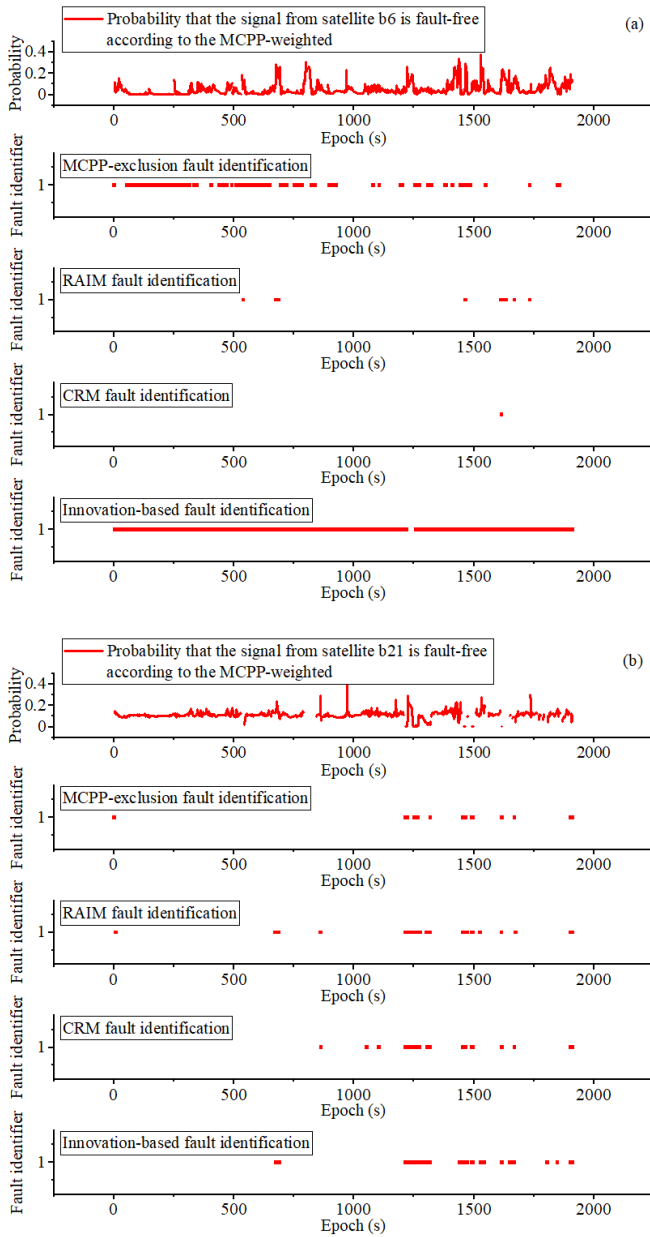


Fig. 4. Fault identification of satellite signals by several methods, (a) fault identification of signals from satellite b6, (b) fault identification of signals from satellite b21. The MCPP-weighted mode gives the normalized probability that the method considers the satellite signal to be error-free in each epoch. The other methods give their respective judgment results on satellite signal failures at each epoch, and each red square represents a successful detection and exclusion of a fault.

other methods can be seen in many places in the results of the positioning errors along time shown in Fig. 6. In particular, around the 1430 epoch, it shows that there are more faulty signals than normal signals, causing the RAIM method to get a false consistency and amplifying the systematic errors. Combining the CDF plot of positioning errors shown in Fig. 7, the purple curve representing the RAIM method starts steeply and then becomes flat. It is in line with the analysis of existing FDE methods in the introduction, i.e., the classical FDE algorithms have a good performance when there is only a single fault in GNSS signals, whereas, when there are multiple faults, solutions based on consistency checking and

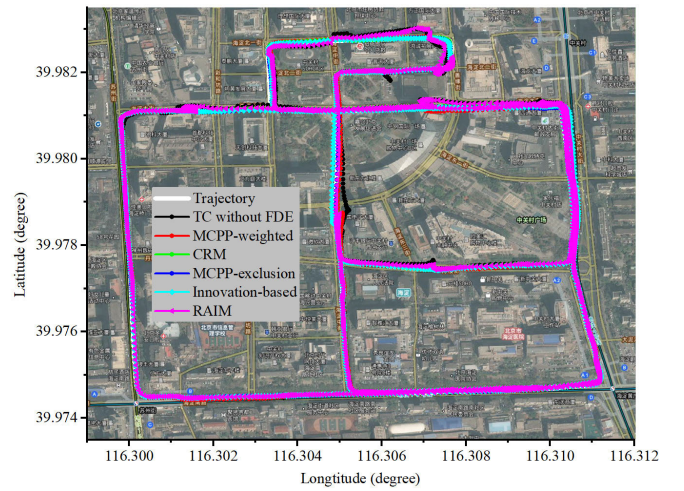


Fig. 5. The positioning error of TC mode with different FDE methods along trajectory.

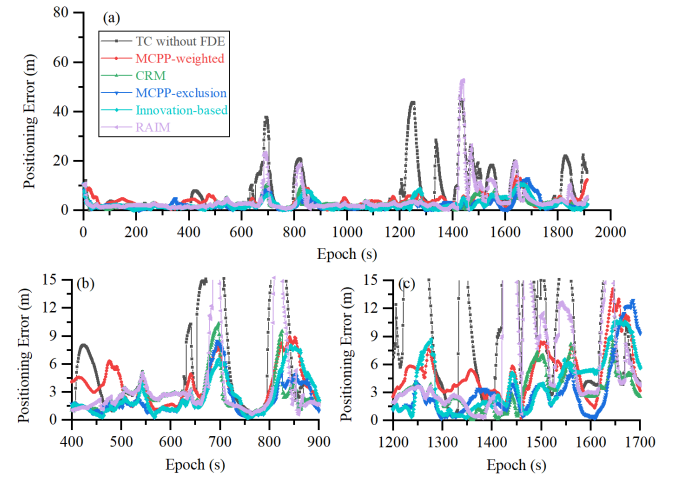


Fig. 6. Positioning error of TC mode with different FDE methods along time, (a) overall positioning error, (b) positioning error details in the first GNSS harsh area, (c) positioning error details in the second GNSS harsh area.

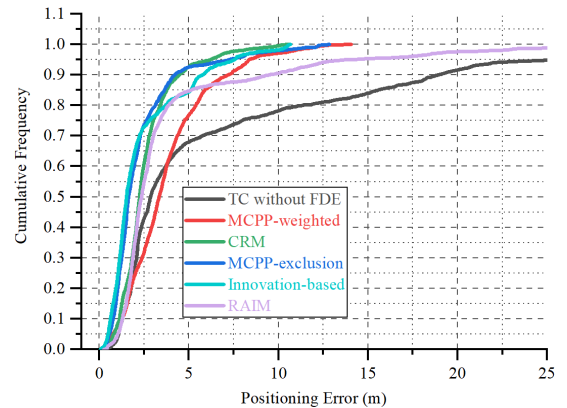


Fig. 7. The positioning error CDFs in TC mode with different FDE method.

not enhancing the redundancy of the system information will not work.

The statistics of standard deviation, 2σ , and 3σ in Table I are all better for the CRM method, which again can be analyzed in combination with the resultant plot of positioning

TABLE I
THE POSITIONING ERROR STATISTICS OF TC MODE WITH DIFFERENT
FDE METHODS IN FULL RANGE

	Mean (m)	Standard Deviation (m)	2σ (m)	3σ (m)	Median (m)	Maximum (m)
TC without FDE	6.95	8.87	17.74	26.61	2.86	46.04
CRM	2.61	1.63	3.25	4.88	2.24	10.63
MCPP-weighted	3.78	2.41	4.82	7.23	3.34	14.09
MCPP-exclusion	2.32	2.11	4.22	6.33	1.63	12.87
Innovation-based	2.41	2.23	4.46	6.68	1.52	10.7
RAIM	4.14	6.09	12.18	18.27	2.36	52.73

errors along time as shown in Fig. 6. In Fig. 6, the errors of all the methods are large around epochs 700, 825, and 1650. Combined with Fig. 2, it is clear that epochs 825 and 1650 pass through the same place. It is presumed that all GNSS signals have faults at these locations. In contrast, the GNSS signals at other locations suffering heavy outlier should have a mix of good and faulty signals. As the vehicle drove past these all-faulty signals locations, the RAIM (although with poor overall performance) and CRM methods, which use purely GNSS system information for FDE, showed the fastest decline in error as normal GNSS signals appeared. The other methods for detecting faulty signals with the help of INS measurements will still be influenced by the results of the previous epoch and maintain a certain degree of inertia. This is unavoidable for systems using IMUs.

Among all the methods, the MCPP-exclusion method has the smallest average positioning error, indicating that the MCPP method does accurately identify the faulty signals. On the other hand, the average positioning error of the MCPP-weighted method is larger. It is due to the fact that the method retains a part of the useful information of the mildly faulty signals while retaining a part of their errors. The overall results of this method are slightly worse than other methods based on error identification and elimination when the system can correctly identify faults. It is especially evident in statistics such as 3σ and median. Also, it is reflected more intuitively on the positioning error CDF plot as shown in Fig. 7. In Fig. 7, the red curve representing the MCPP-weighted method is steep but shifted to the right compared to the other three methods, indicating that this method has a slightly larger overall positioning error but does not suffer from extreme error situations.

As analyzed at the beginning of Section IV-A, the signals of systems operating in the real world contain many unobservable errors, and there is no way to determine the so-called true value. So, whether a signal is fault depends on how much it contributes to the positioning error. MCPP's utilization of fault signals is essentially a tradeoff by giving small weights to possibly faulty signals thereby adding extra redundancy to the system. MCPP allows the system performance to remain unaffected when signals are good, and ensures that the system does not diverge (due to wrong consistency excluding correct

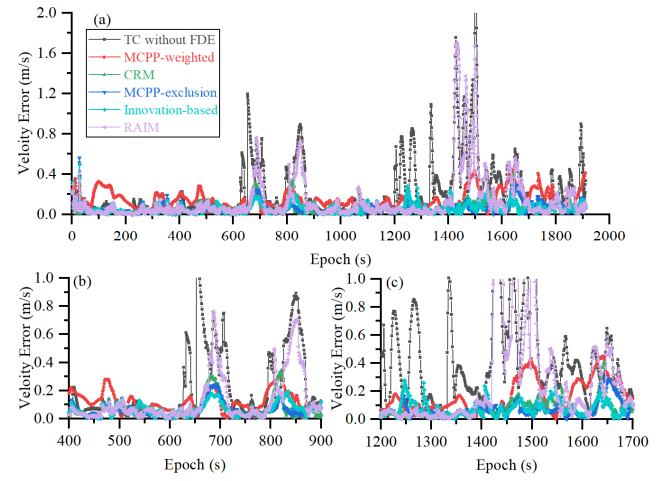


Fig. 8. Velocity error of TC mode with different FDE methods along time, (a) overall velocity error over time, (b) velocity error details in the first GNSS harsh area, (c) velocity error details in the second GNSS harsh area.

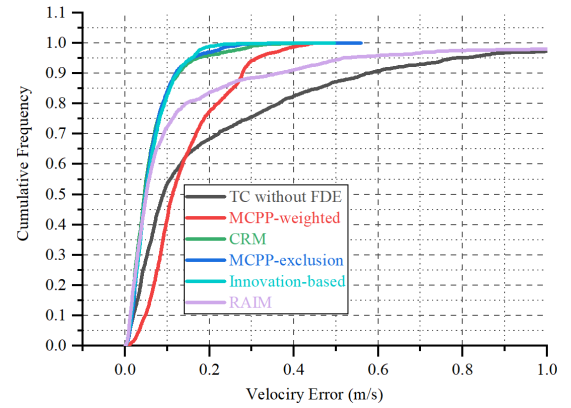


Fig. 9. The velocity error CDFs in TC mode with different FDE method.

signals) when signals are poor. The MCPP method obtains the worst performance when the system is significantly affected but good signals are still in the majority and the elimination method correctly eliminates the wrong signals.

However, as mentioned in the introduction, although the weight-based method retains a certain amount of error, it retains all the useful information and does not need to consider the issue of threshold setting, which makes this method more universal and applicable to all scenarios when applied. This is better demonstrated in the analysis in the next section.

The velocity error statistics for these methods in TC mode are shown in Table II. Their distributed velocity performance along time in TC mode are shown in Fig. 8. Their velocity error CDF are shown in Fig. 9.

When it comes to velocity several methods show a similar situation as in positioning, with only one point that deserves additional attention. When measuring velocity, CRM, MCPP-exclusion, and innovation-based method demonstrated similar performance, all accurately identifying false signals. Although the MCPP-weighted method also demonstrates good performance, the gap with the other three methods is larger than their gap in positioning. This is due to the fact that the GNSS

TABLE II
THE VELOCITY ERROR STATISTICS OF TC MODE WITH DIFFERENT FDE
METHODS IN FULL RANGE

	Mean (m/s)	Standard Deviation (m/s)	2σ (m/s)	3σ (m/s)	Median (m/s)	Maximum (m/s)
TC without FDE	0.22	0.29	0.58	0.88	0.09	2.61
CRM	0.06	0.06	0.12	0.18	0.05	0.4
MCPP-weighted	0.14	0.09	0.18	0.27	0.11	0.45
MCPP-exclusion	0.06	0.05	0.1	0.15	0.05	0.56
Innovation-based	0.06	0.05	0.09	0.14	0.05	0.5
RAIM	0.13	0.24	0.47	0.71	0.05	1.7

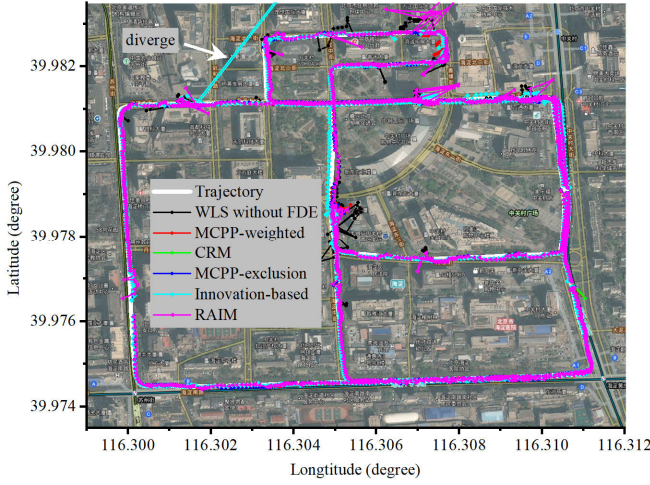


Fig. 10. The positioning error of WLS mode with different FDE methods along trajectory.

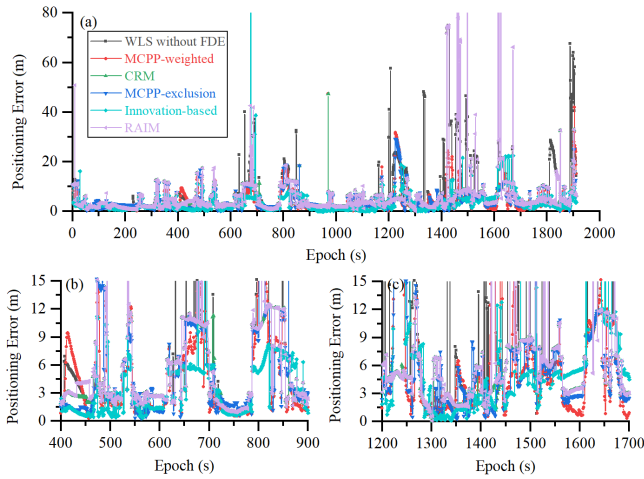


Fig. 11. Positioning error of WLS mode with different FDE method, (a) overall positioning error over time, (b) positioning error details in the first GNSS harsh area, (c) positioning error details in the second GNSS harsh area.

measurements are not directly involved in positioning, velocity measurement and attitude determination, but only serve as the correction of INS errors by the filter, during the operation of the tightly coupled system. For the velocity section, since doppler measurements are inherently insensitive to the effects

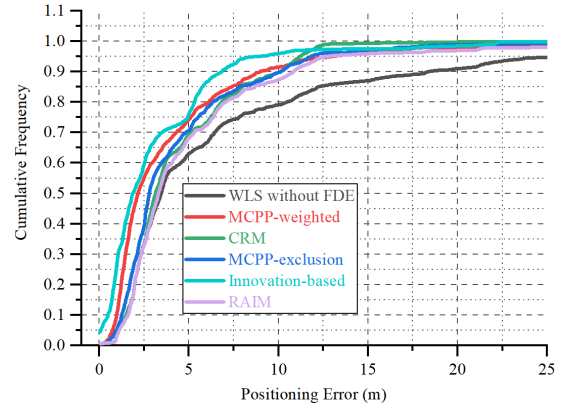


Fig. 12. The positioning error CDFs in WLS mode with different FDE methods along trajectory.

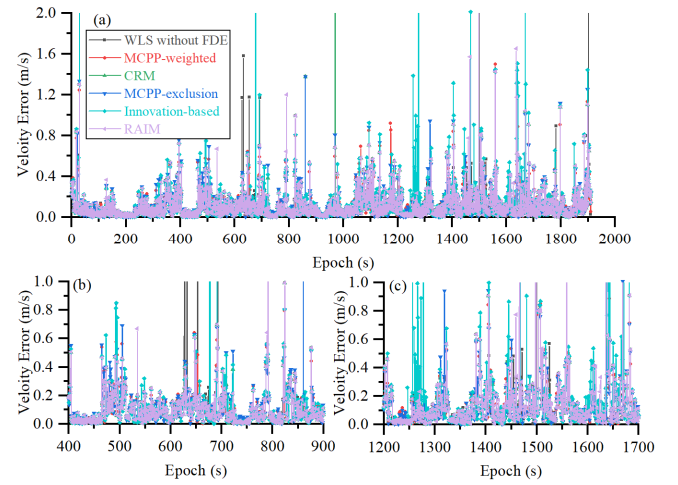


Fig. 13. Velocity error of WLS mode with different FDE methods along trajectory, (a) overall velocity error over time, (b) velocity error details in the first GNSS harsh area, (c) velocity error details in the second GNSS harsh area.

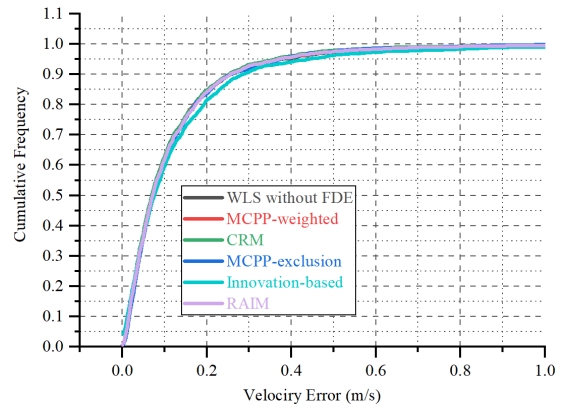


Fig. 14. The velocity error CDFs for several methods in WLS mode.

of reflected signals, it is less likely that the level of error in measurements from different satellites differ significantly, as is the case with positioning. It is easier for the system to recognize when there is an error in the doppler information, and the correction of the INS velocity is accurate when the fault is eliminated. In this case, if the MCPP-weighted scheme

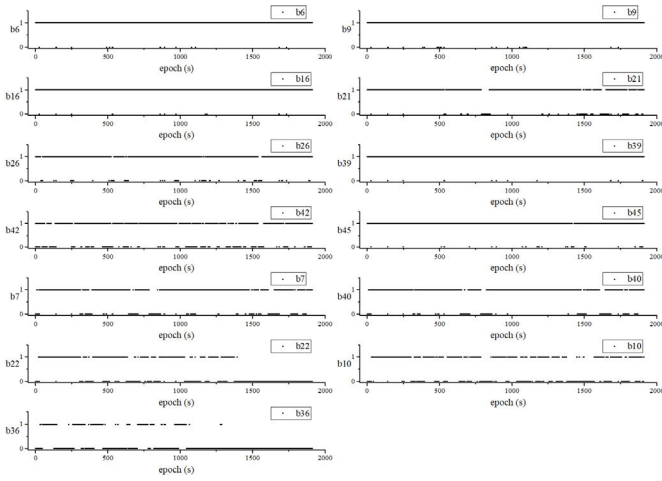


Fig. 15. Original satellite visibility.

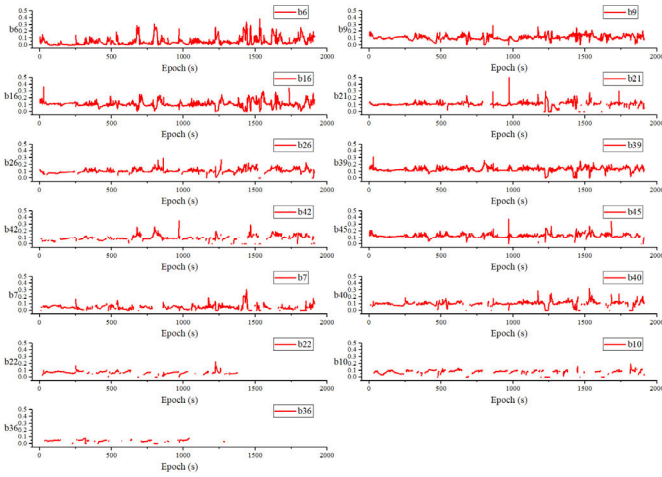


Fig. 16. The probability that the signal is error free according to the MCP-weighted mode. Blank indicates that the satellite is not visible.

is used, the additional information with the error term has a negative effect.

C. Multi-Channel Parallel Processing Method in Cooperative Positioning

With the development of internet of things (IOT) in the field of transport, new concepts such as vehicle-to-road and vehicle-to-vehicle cooperation are being applied in navigation processes. Taking advantages of vehicle-to-vehicle communications, a vehicle can receive extra sensor data shared by neighboring vehicles/infrastructures to achieve cooperative positioning. Most of the existing cooperation positioning methods assume the availability of inter-vehicle ranging. However, due to the complexity of such a system and the difficulty of guaranteeing real-time performance, we only consider broadcasting its FDE information in real time through the head vehicle or the ground facility in the cluster, thus enhancing the navigation performance of the surrounding vehicles. This section verifies the performance of GNSS signals purified by several FDE solutions for single-point positioning, so that their performance when used for cooperative positioning can be obtained.

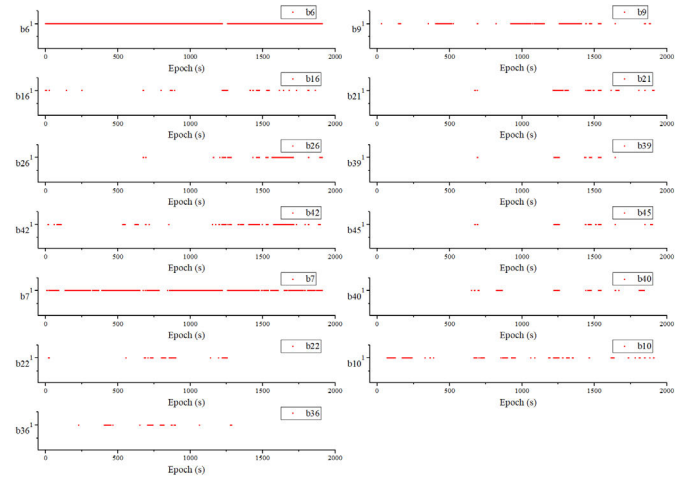


Fig. 17. FDE results of the Innovation-based method. Each red square represents a successful detection and exclusion of a fault. The threshold here is chosen to be 1.6 based on the exhaustive method.

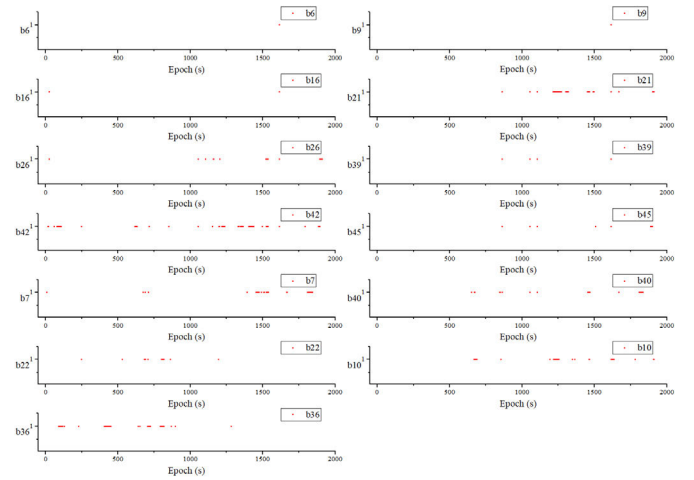


Fig. 18. FDE results of the CRM method. Each red square represents a successful detection and exclusion of a fault.

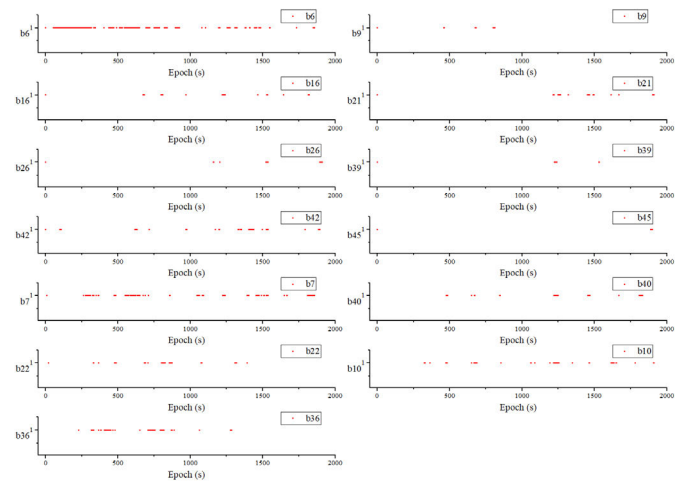


Fig. 19. FDE results of the MCP-weighted exclusion method. Each red square represents a successful detection and exclusion of a fault. The probability threshold here is chosen to be 0.041 based on the exhaustive method.

The positioning error statistics for these methods in WLS positioning mode are shown in Table III. Their distributed

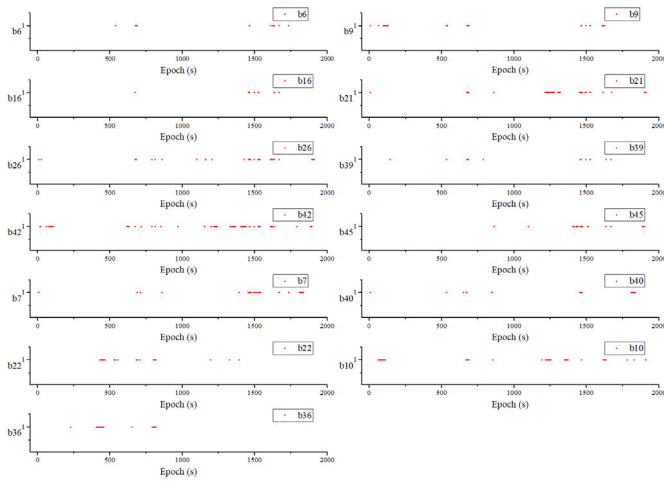


Fig. 20. FDE results of the RAIM method.

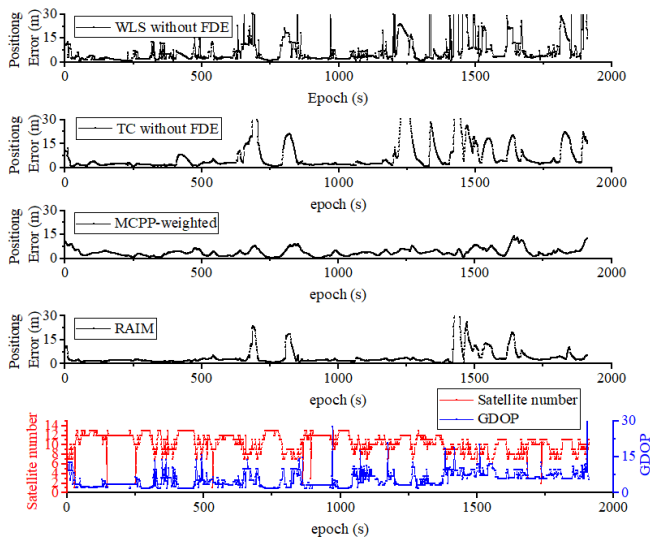


Fig. 21. Distribution of the number of visible satellites, DOP, and positioning errors.

TABLE III

THE POSITIONING ERROR STATISTICS OF WLS MODE WITH DIFFERENT FDE METHODS IN FULL RANGE

	N missing ^a	Mean (m)	Standard Deviation (m)	2 σ (m)	3 σ (m)	Median (m)	Maximum (m)
WLS without FDE	10	7.34	9.97	19.94	29.92	3.47	74.96
CRM	14	4.51	3.53	7.06	10.59	3.15	47.37
MCPP-weighted	10	4.09	4.77	9.55	14.32	2.17	47.37
MCPP-exclusion	11	4.45	4.15	8.3	12.45	2.8	29.06
Innovation-based	81	6.05	82.42	164.84	247.26	2.03	2504.07
RAIM	14	5.92	10.91	21.83	32.74	3.32	250.86

^a N missing is when the number of purified GNSS signals is less than 4.

positioning performance along trajectory and time in WLS positioning mode are shown in Fig. 10 and Fig. 11 respectively. Their positioning error CDF are shown in Fig. 12.

The positioning errors of WLS compared to the results in the tightly coupled mode in the previous section show

TABLE IV
THE VELOCITY ERROR STATISTICS OF WLS MODE WITH DIFFERENT FDE METHODS IN FULL RANGE

	N missing	Mean (m/s)	Standard Deviation (m/s)	2 σ (m/s)	3 σ (m/s)	Median (m/s)	Maximum (m/s)
WLS without FDE	10	0.14	0.54	1.08	1.62	0.07	21.51
CRM	14	0.13	0.51	1.02	1.53	0.07	21.51
MCPP-weighted	10	0.13	0.51	1.02	1.53	0.07	21.51
MCPP-exclusion	11	0.12	0.14	0.28	0.42	0.07	1.43
Innovation-based	81	0.16	0.64	1.29	1.93	0.08	20.36
RAIM	14	0.12	0.23	0.47	0.7	0.07	7.95

the most severe deterioration in the positioning performance of the WLS and RAIM methods, and the innovation-based method shows divergence points. In the integrated navigation, the GNSS results are not output separately but are only used to correct the INS errors. During the operation of the integrated navigation filter, when a part of the errors is retained in the measurements, these errors affect the process of repairing the INS errors by the system. So, the method based on error elimination in integrated navigation is better. Whereas in the WLS positioning process, only GNSS information is available. It would be more important to keep some more GNSS information when retaining errors than when there is INS present.

On the other hand, the integrated system is more tolerant to threshold selection due to the presence of INS. The WLS mode also needs to be more careful about the selection of the threshold value. A loose selection of thresholds, while allowing the system to work more consistently, can make the method insufficiently capable of identifying faults. In contrast, a strict selection of the threshold value may result in a poor geometric configuration of the remaining satellites, which will also result in a degradation of the positioning performance, or even in a situation where the number of available satellites is insufficient for positioning. Since the thresholds here are determined based on the average positioning error of the TC modes. It is conceivable that although the MCPP-exclusion method is stable and not as diffuse as the Innovation-based method, they are essentially the same class of methods (including CRM schemes), and both of them need to fine-tune the thresholds in order to obtain the best performance, whereas the MCPP-weighted method is more universal and more capable of handling a wider range of scenarios.

The velocity error statistics for these methods in WLS mode are shown in Table IV. Their distributed velocity performance along time in WLS mode are shown in Fig. 13. Their velocity error CDF are shown in Fig. 14.

In terms of velocity, as mentioned in the previous section, there is no enhancement to the velocity results from identifying and removing these reflections as the Doppler measurements are inherently insensitive to the effects of reflected signals, which are the main source of error. The curve representing the Innovation-based scheme is lower than the other curves because of the divergence of the innovation-based method at

the place as shown in Fig. 10. So, the improvement in velocity results in TC mode is mainly due to the use of IMU devices.

V. CONCLUSION

In this paper, a multi-channel parallelism FDE method is proposed. The method inputs measurements from each satellite into separate channels for calculation. During the calculation, in each sub-model, the difference between the measurement of a satellite and its INS estimate version is compared. These differences are normalized by a multiple model interaction process to derive the probability that each satellite is error free. These probabilities can be used as weights for the optimal output or as measures for identifying false signals, corresponding to MCPP-weighted mode and MCPP-exclusion mode respectively.

The experimental results in the Zhongguancun area of Beijing show that the MCPP method can effectively identify GNSS faulty values. The performance of its exclusion mode is competitive with state-of-the-art exclusion methods when the thresholds are carefully selected. The weighted mode as an optimal processing method can optimize the output in any environment and in any operating conditions without considering the threshold selection.

After eliminating the fault caused by the GNSS, future work will focus on the fault caused by the low-cost INS component of integrated navigation system.

APPENDIX

Fig. 15 to 20 in this section show the availability of each satellite corresponding to each method in part B and part C of section IV.

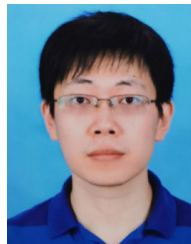
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